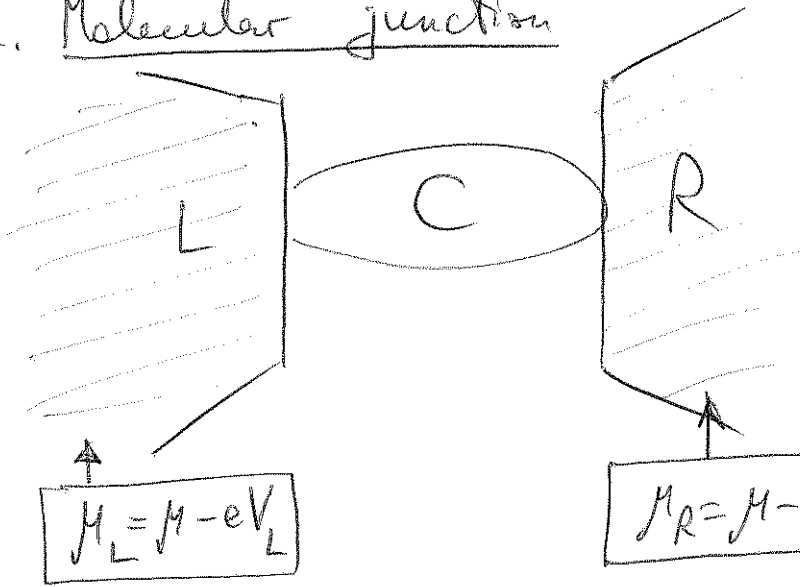
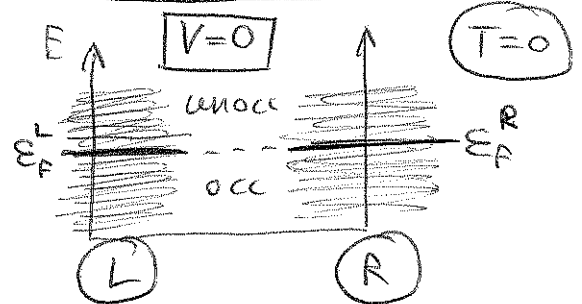


Quantum conductance

1. Molecular junction

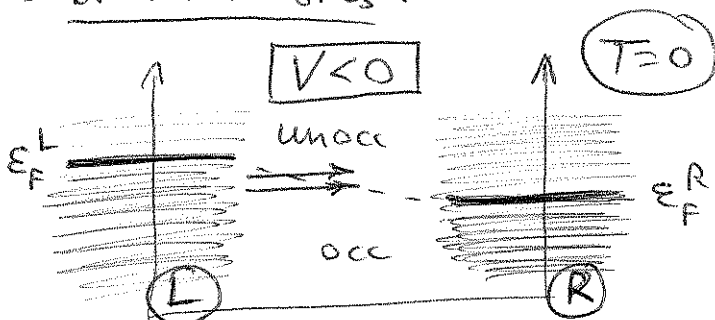


• Zero bias:

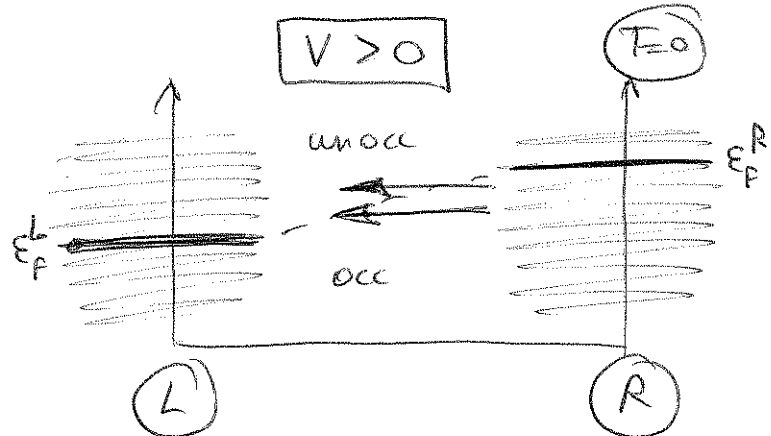


Fermi energies are aligned. If two materials are different, there will be a contact potential difference (CPD)

• Non-zero bias:



e^- flow from the surface within $\epsilon_F^L - eV < E < \epsilon_F^L$ energy interval, $L \rightarrow R$



e^- flow from the $R \rightarrow L$

In both cases the Fermi levels are displaced by eV with respect to each other.

2. The Hamiltonian

$$H = H_L + H_R + H_I + H_C$$

$$\sum_n \epsilon_n a_n^\dagger a_n \quad \text{central region}$$

(1)

left lead $\sum_e (\epsilon_e - \mu_e) a_e^\dagger a_e$

right lead $\sum_r (\epsilon_r - \mu_r) a_r^\dagger a_r$

$$\sum_{en} (V_{en} a_e^\dagger a_n + V_{ne} a_n^\dagger a_e)$$

$$+ \sum_{rn} (V_{rn} a_r^\dagger a_n + V_{nr} a_n^\dagger a_r)$$

interaction between regions

There is no "interaction" between R and L directly.

Also, operators of different regions (L, C, R) commute; we used site-representation, as our functions are localised along the dissection of the junction. $L \rightarrow C \rightarrow R$.

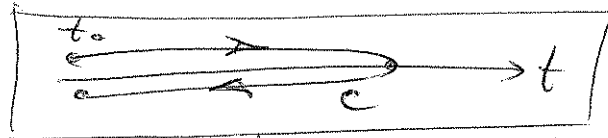
Another form:

$$\hat{H} = \sum_{ij} H_{ij} a_i^\dagger a_j \quad (2)$$

$$H = \begin{pmatrix} H_L & V_{LC} & 0 \\ V_{CL} & H_C & V_{CR} \\ 0 & V_{RC} & H_R \end{pmatrix}, \quad \begin{matrix} V_{LC} = V_{CL}^T \\ V_{CR} = V_{RC}^T \end{matrix} \quad (3)$$

$$\{a_i, a_j^\dagger\} = \delta_{ij}; \quad \{a_i, a_j\} = \{a_i^\dagger, a_j^\dagger\} = 0 \quad (4)$$

3. The Green's function:



$$G_{ij}^<(t, t') = -i \langle T_C c_{ji}(t) c_{ji}^\dagger(t') \rangle = \theta_{tt'} G^> + \theta_{t't} G^< \quad (5)$$

$$c_{ji}(t) = e^{iH_C t} c_i e^{-iH_C t} \rightarrow \text{in Heisenberg picture}$$

This GF is a coefficient in the expansion of ψ -based GF:

$$G(1, 2) = -i \langle T_C \psi_{\mathcal{H}}(1) \psi_{\mathcal{H}}^\dagger(2) \rangle =$$

$$\psi_{\mathcal{H}}(1) = \sum_i \chi_i(x_1) a_{ji}(t_1), \text{ etc.}$$

$$\hookrightarrow G(1, 2) = \sum_{ij} \chi_i(x_1) \chi_j^*(x_2) \underbrace{G_{ij}^<(t_1, t_2)}_{\text{only depends on times}}$$

Since $\hat{H} = \int \psi^\dagger(x) \hat{h}(x) \psi(x) dx$ with the $\hat{h}$ assumed in H_{ij} in (3), we can immediately use our previous results for the EoM for G_{ij} :

$$i \partial_t G_{ij}^<(t, t') = \delta_{tt'} \delta_{ij} + \sum_k H_{ik} G_{kj}^<(t, t') \quad (6)$$

$$-i\partial_{t'} G_{ij}^L(t, t') = \delta_{tt'} \delta_{ij} + \sum_k G_{ik}^L(t, t') H_{kj} \quad (7)$$

4. Green functions of the regions (separately)

(i) Left lead

$$i\partial_t g_{ij}^L(t, t') = \sum_k \cancel{H_{ik}^L} H_{ik}^L g_{kj}^L(t, t') = \delta_{tt'} \delta_{ij} \quad (8)$$

or $(i\partial_t \mathbb{1}_{LL} - H_{LL}) g^L(t, t') = \delta_{tt'} \mathbb{1}_{LL}$ (matrix form) (9)

We can find components $g^<$ and $g^>$ of g^L :

$$g_{ij}^{<(L)}(t, t') = i\delta_{ij} f(\bar{\epsilon}_i) e^{-i\bar{\epsilon}_i(t-t')} \equiv \delta_{ij} g_i^{<(L)}(t, t') \quad (10)$$

$$g_{ij}^{>(L)}(t, t') = -i\delta_{ij} (1 - f(\bar{\epsilon}_i)) e^{-i\bar{\epsilon}_i(t-t')} \equiv \delta_{ij} g_i^{>(L)}(t, t') \quad (11)$$

Their FT:

$$g_{ij}^{<(L)}(\omega) = \int d\tau e^{i\omega\tau} g_{ij}^{<(L)}(\tau) \stackrel{t-t'}{\rightarrow} \equiv \delta_{ij} 2\pi i f(\bar{\epsilon}_i) \delta(\omega - \bar{\epsilon}_i) \quad (12)$$

$$g_{ij}^{>(L)}(\omega) = -2\pi i \delta_{ij} (1 - f(\bar{\epsilon}_i)) \delta(\omega - \bar{\epsilon}_i) \quad (13)$$

(ii) The same expressions can be written for the $g_{ij}^R(t, t')$ for the right lead. And for $g_{ij}^C(t, t')$ associated with the central region C, for the latter we need actually $g^{R(C)}$ and $g^{A(C)}$:

$$g_{ij}^{R(C)}(t, t') = -i\delta_{ij} \theta(t-t') e^{-i\bar{\epsilon}_i(t-t')} \quad (14)$$

$$g_{ij}^{R(C)}(\omega) = \delta_{ij} (\omega^+ - \bar{\epsilon}_i)^{-1}, \quad \omega^+ = \omega + i\delta \quad (15)$$

$$g_{ij}^{A(C)}(\omega) = \delta_{ij} (\omega^- - \bar{\epsilon}_i)^{-1}, \quad \omega^- = \omega - i\delta \quad (16)$$

In practice, there is some $\mu(x)$ (x -dissection $L \rightarrow C \rightarrow R$) due to

$$\mu(x) = \mu - e \underbrace{V(x)}$$

↳ electrostatic potential across the junction.

We consider a simplified picture here,

More explicitly for the R,A parts:

$$\begin{cases} g_{cc}^R(\omega) = \mathbb{1}_c \left[\mathcal{P} \frac{1}{\omega - \epsilon_c} - i\pi \delta(\omega - \epsilon_c) \right] \\ g_{cc}^A(\omega) = \mathbb{1}_c \left[\mathcal{P} \frac{1}{\omega - \epsilon_c} + i\pi \delta(\omega - \epsilon_c) \right] \end{cases} \quad (17)$$

Similarly for R,A parts of $g^L(\omega)$ and $g^R(\omega)$.

5. EoM and extracting the "perturbation"

$$\mathcal{H} = H^0 + W, \quad H^0 = \begin{pmatrix} H_L & 0 & 0 \\ 0 & H_c & 0 \\ 0 & 0 & H_R \end{pmatrix}, \quad W = \underbrace{\begin{pmatrix} 0 & V_{LC} & 0 \\ V_{CL} & 0 & V_{CR} \\ 0 & V_{RC} & 0 \end{pmatrix}}_{\text{interactions } \cancel{L \leftrightarrow R} \quad L \leftrightarrow C \leftrightarrow R} \quad (18)$$

Rewrite (6):

$$(i\partial_t - H^0) G(t, t') = \delta_{tt'} \mathbb{1} + W G(t, t') \quad (19)$$

and then write it for the blocks KK' ($K, K' = L, c \text{ or } R$):

$$\boxed{(i\partial_t - H_K^0) G_{KK'}^I(t, t') = \delta_{tt'} \delta_{KK'} + \sum_{M \in L, c, R} W_{KM} G_{MK'}^I(t, t')} \quad (20)$$

↑
diagonal

Introduce the $\int_{t'}^{t} d\tau$ in the LHS:



$$\text{LHS} = \int_C d\tau \underbrace{(i\partial_t - H_K^0) \delta_{\tau t}}_{g_K^{\mathbb{E}}(t, \tau)^{-1} \text{ (see (9))}} G_{KK'}^I(\tau, t') \equiv \int_C d\tau g_K(t, \tau)^{-1} G_{KK'}^I(\tau, t') \quad (21)$$

Multiply both sides of (20) on $\int_C g(t_1, t) dt$

↳ ~~$G_{kk}(t, t') = g_{kk}$~~ In matrix notation:

$$\int_C d\tau g(t, \tau)^{-1} G(\tau, t') = \delta_{t, t'} \mathbb{1} + W G(t, t')$$

$$\hookrightarrow \boxed{G(t_1, t') = g(t_1, t') \mathbb{1} + \int_C g(t_1, t) W G(t, t') dt} \quad (22)$$

where

$$g(t, t') = \begin{pmatrix} g_L & 0 & 0 \\ 0 & g_c & 0 \\ 0 & 0 & g_R \end{pmatrix} \quad (23)$$

• In particular, consider the LC component of (22):

$$G_{LC}^R(t, t') = \delta_{LC} g(t, t') + \int_C g_{LK}(t, \tau) W_{KK'} G(\tau, t') d\tau \quad (23^*)$$

(summations over $k, k' = L, C, R$ are implied)

Using Langreth rules (t_0 is taken at $-\infty$):

$$G_{LC}^R(t, t') = \int_{-\infty}^{\infty} d\tau g_L^R(t, \tau) W_{LC} G_{CC}^R(\tau, t')$$

(there is only one W_{LK} component of W which is W_{LC})

$$\hookrightarrow \boxed{G_{LC}^R(\omega) = g_L^R(\omega) W_{LC} G_{CC}^R(\omega)} \quad (24)$$

Also, in the same way for $G_{RC}^R(\omega)$:

$$\boxed{G_{RC}^R(\omega) = g_R^R(\omega) W_{RC} G_{CC}^R(\omega)} \quad (25)$$

Equations for the A-component look the same.

• Now consider $kk' = CC$ component of (22):

$$G_{CC}(t, t') = g_C(t, t') + \int_C d\tau g_C(t, \tau) [W_{CL} G_{LC}(\tau, t') + W_{CR} G_{RC}(\tau, t')] \quad (26)$$

Use Langreth rules:

$$G_{cc}^R(t, t') = g_c^R(t, t') + \int_{-\infty}^{\infty} d\tau g_c^R(t, \tau) [W_{CL} G_{LC}^R(\tau, t') + W_{CR} G_{RC}^R(\tau, t')]$$

FT of which gives:

$$G_{cc}^R(\omega) = g_c^R(\omega) + g_c^R(\omega) \left[\underbrace{W_{CL} G_{LC}^R(\omega)}_{\text{take from (24)}} + \underbrace{W_{CR} G_{RC}^R(\omega)}_{\text{use (25)}} \right] \quad (27)$$

$$= g_c^R(\omega) + g_c^R(\omega) \left[W_{CL} g_L^R(\omega) W_{LC} + W_{CR} g_R^R(\omega) W_{RC} \right] G_{cc}^R(\omega)$$

$\Sigma_{cc}^R(\omega) \rightarrow \text{leads "self-energy"}$

so,

$$\boxed{G_{cc}^R(\omega) = g_c^R(\omega) + g_c^R(\omega) \Sigma_{cc}^R(\omega) G_{cc}^R(\omega)} \quad (28)$$

Looks like the Dyson equation. Here

$$\Sigma_{cc}^R(\omega) = W_{CL} g_L^R(\omega) W_{LC} + W_{CR} g_R^R(\omega) W_{RC}$$

$$\equiv V_{CL} g_L^R V_{LC} + V_{CR} g_R^R V_{RC} \quad (29)$$

The eq. for $G_{cc}^A(\omega)$ looks similarly to (28).

• We also need $G_{cc}^<(\omega)$. Project (22) on $\omega < 0$ and use Langreth rules:

$$\cancel{G_{cc}^<(t, t') = g_c^<(t, t') + \int_{-\infty}^{\infty} d\tau g_c^<(t, \tau) [W_{CL} G_{LC}^<(\tau, t') + W_{CR} G_{RC}^<(\tau, t')]} + \int_{-\infty}^{\infty} d\tau g_c^R(t, \tau) [W_{CL} G_{LC}^<(\tau, t') + W_{CR} G_{RC}^<(\tau, t')]$$

$$G_{cc}^<(t, t') = g_c^<(t, t') + \int_{-\infty}^{\infty} d\tau \left\{ g_c^R(t, \tau) W G_{cc}^<(\tau, t') + g_c^<(t, \tau) W G_{cc}^A(\tau, t') \right\} \quad (30)$$

or

$$G_{cc}^<(\omega) = g_c^<(\omega) + g_c^R(\omega) W G_{cc}^<(\omega) + g_c^<(\omega) W G_{cc}^A(\omega) \quad (31)$$

In particular, consider the CC-component (omit ω for simplicity).

$$G_{cc}^< = g_c^< + g_c^R [W_{cL} G_{Lc}^< + W_{cR} G_{Rc}^<] + g_c^< [W_{cL} G_{Lc}^A + W_{cR} G_{Rc}^A] \quad (32)$$

G_{Lc}^A, G_{Rc}^A can be taken from the A-analogs of (24), (25); however, it is clear that we also need $G_{Lc}^<$ and $G_{Rc}^<$.

• Hence project (23)* on < component and use the Langreth rules:

$$G_{Lc}^<(t, t') = \int_{-\infty}^{\infty} d\tau \left[g_L^R(t, \tau) W_{Lc} G_{cc}^R(\tau, t') + g_L^<(t, \tau) W_{Lc} G_{cc}^A(\tau, t') \right] \quad (33)$$

which after the FT is;

$$\boxed{G_{Lc}^< = g_L^R W_{Lc} G_{cc}^< + g_L^< W_{Lc} G_{cc}^A} \quad (34)$$

and similarly for $G_{Rc}^<$. Substitute this into (32), also use (24), (25):

$$\begin{aligned} G_{cc}^< &= g_c^< + g_c^R \left\{ W_{cL} (g_L^R W_{Lc} G_{cc}^< + g_L^< W_{Lc} G_{cc}^A) + \right. \\ &\quad \left. + W_{cR} (g_R^R W_{Rc} G_{cc}^< + g_R^< W_{Rc} G_{cc}^A) \right\} \\ &+ g_c^< \left\{ W_{cL} g_L^A W_{Lc} G_{cc}^A + W_{cR} g_R^A W_{Rc} G_{cc}^A \right\} \\ &= g_c^< + \cancel{g_c^R W_{cL}} g_c^R \left\{ [W_{cL} g_L^R W_{Lc} + W_{cR} g_R^R W_{Rc}] G_{cc}^< \right. \\ &\quad \left. + [W_{cL} g_L^< W_{Lc} + W_{cR} g_R^< W_{Rc}] G_{cc}^A \right\} \\ &+ g_c^< [W_{cL} g_L^A W_{Lc} + W_{cR} g_R^A W_{Rc}] G_{cc}^A \end{aligned}$$

Expressions in [...] are various incarnations of the leads self-energy (29):

$$G_{cc}^<(\omega) = g_c^<(\omega) + g_c^R(\omega) \left[\sum_{cc}^R(\omega) G_{cc}^<(\omega) + \sum_{cc}^<(\omega) G_{cc}^A(\omega) \right] + g_c^<(\omega) \sum_{cc}^A(\omega) G_{cc}^A(\omega) \quad (35)$$

- This has a similar form to the Keldysh equation from Lecture 5. Indeed, compare with (5.21). Also, we have another Keldysh equation for the $R(A)$ component which is analogous to our formula (28). These similarities explain why $\Sigma(\omega)$ is also called 'self-energy'.

Also, using this similarity, we see that in the steady state:

$$G_{cc}^<(\omega) = G_{cc}^R(\omega) \sum_{cc}^<(\omega) G_{cc}^A(\omega) \quad (36)$$

as the 1st term in the Keldysh formula $\rightarrow 0$.

So, we now have:

$$G_{cc}^<(\omega) \rightarrow \text{eq. (36)}$$

$$G_{cc}^A(\omega), G_{cc}^R(\omega) \rightarrow \text{eq. (28). This is self-contained.}$$

Solving for $G_{cc}^{A/R}(\omega)$ allows calculating

$G_{cc}^<(\omega)$ from (36):

$$G_{cc}^{A/R}(\omega) = (1_{cc} - g_c^R(\omega) \sum_{cc}^R(\omega))^{-1} g_c^R(\omega) \quad (37)$$

as the region C is finite, i.e. here we have finite multiplication, and inversion of a finite matrix.

6. The current

"From" the left electrode (or "to"):

$$I_L = -e \langle \dot{N}_L \rangle, \quad N_L = \sum_{e \in L} a_e^\dagger a_e - e \text{ number operator} \quad (38)$$

Here $\dot{N}_L = \frac{i}{\hbar} [\mathcal{H}, N_L]$ (omit $\hbar$ in the following)

Calculating the commutator:

$$[\mathcal{H}, N_L] = [H_L + \underbrace{H_C + H_R + H_T}_{\substack{\text{irrelevant} \\ \text{as belong to} \\ \text{different regions}}}, N_L] = \underbrace{[H_L, N_L]}_{\substack{\text{they commute} \\ \text{non-zero}}} + \underbrace{[H_T, N_L]}_{\substack{\text{the only} \\ \text{non-zero}}}$$

~~$[\mathcal{H}, N_L] = \dots$~~ It is easier to do the calculation with the full $\hat{H}$:

$$[\mathcal{H}, N_L] = \sum_{i,j} H_{ij} \sum_{k \in L} [a_i^\dagger a_j, a_k^\dagger a_k]$$

$$[a_i^\dagger a_j, a_k^\dagger a_k] = \underbrace{a_i^\dagger a_j a_k^\dagger a_k}_{\delta_{jk} - a_k^\dagger a_j} - \underbrace{a_k^\dagger a_k a_i^\dagger a_j}_{\delta_{ki} - a_i^\dagger a_k} =$$

$$= \delta_{jk} a_i^\dagger a_k - \delta_{ki} a_k^\dagger a_j - \cancel{a_i^\dagger a_k^\dagger a_j a_k} + \underbrace{\cancel{a_k^\dagger a_i^\dagger a_k a_j}}_{\substack{-a_i^\dagger a_k^\dagger - a_j a_k \\ + a_i^\dagger a_k^\dagger a_j a_k}}$$

$$= \delta_{jk} a_i^\dagger a_k - \delta_{ki} a_k^\dagger a_j \quad (39)$$

$$\hookrightarrow [\mathcal{H}, N_L] = \sum_{i,j} \sum_{k \in L} H_{ij} (\delta_{jk} a_i^\dagger a_k - \delta_{ki} a_k^\dagger a_j)$$

$$= \sum_{k \in L} \sum_i (H_{ik} a_i^\dagger a_k - H_{ki} a_k^\dagger a_i) \quad (40)$$

If $i \in L$, then

$$(\dots) = \sum_{k \in L} (H_{ik} a_i^\dagger a_k - H_{ki} a_k^\dagger a_i) = \sum_{k \in L} (H_{ik} a_i^\dagger a_k - H_{ik} a_i^\dagger a_k) = 0$$

after $i \leftrightarrow k$ in the 2nd sum, i.e. $i \in C$ only in (40):

$$[H, N_L] = \sum_{j \in L} \sum_{i \in C} (H_{ij} a_i^\dagger a_j - H_{ji} a_j^\dagger a_i) \quad (41)$$

Hence,

$$Y_L = -ie \sum_{i,j \in L} [H_{ij} \langle a_i^\dagger a_j \rangle - H_{ji} \langle a_j^\dagger a_i \rangle] \quad (42)$$

where $\langle a_i^\dagger a_j \rangle = \frac{\text{Tr}[e^{-\beta H} a_i^\dagger a_j]}{\text{Tr}[e^{-\beta H}]} = \frac{\text{Tr}[e^{-\beta H} a_{ji}^\dagger(t) a_{ji}(t)]}{\text{Tr}[e^{-\beta H}]}$

Since $\text{Tr}[e^{-\beta H} a_{ji}^\dagger(t) a_{ji}(t)] = \text{Tr}[e^{-\beta H} (e^{iHt} a_i^\dagger e^{-iHt}) (e^{iHt} a_j e^{-iHt})]$
 $= \text{Tr}[e^{-\beta H} e^{iHt} a_i^\dagger a_j e^{-iHt}] \xrightarrow{\text{cyclic invariance of the trace}}$

Hence,

$$\langle a_i^\dagger a_j \rangle = \langle a_{ji}^\dagger(t) a_{ji}(t) \rangle = -i G_{ji}^<(t, t) \quad (43)$$

and

$$\langle a_j^\dagger a_i \rangle = -i G_{ij}^<(t, t) \quad (44)$$

so that

$$Y_L = +ie \sum_{i,j \in L} [H_{ij} G_{ji}^<(t, t) - H_{ji} G_{ij}^<(t, t)]$$

$$= H_{ij}^* - G_{ji}^<(t, t)^*$$

$$= -2e \sum_{\substack{i \in C \\ j \in L}} \text{Re}[H_{ij} G_{ji}^<(t, t)] = -2e \text{Re}[\underbrace{H_{LL} G_{LL}^<(t, t)}_{\text{trace over } C}] \quad (45)$$

Also, $G_{LC}^<(t, t) = \int \frac{d\omega}{2\pi} e^{-i\omega \cdot 0} G_{LC}^<(\omega) = \int \frac{d\omega}{2\pi} G_{LC}^<(\omega) \quad (46)$

$$\hookrightarrow \Sigma_{ij}^{R(L)}(\omega) = \sum_{\alpha, \beta \in L} V_{i\alpha} V_{\alpha\beta} V_{\beta j} \left[\mathcal{P} \frac{1}{\omega - \bar{\epsilon}_{\alpha\beta}} - i\pi \delta(\omega - \bar{\epsilon}_{\alpha\beta}) \right]$$

$$= \sum_{\beta} \int d\epsilon \underbrace{\rho_{\beta}^L(\epsilon)}_{\text{for the given chemical potential } \mu_L} V_{i\beta}(\epsilon) V_{\beta j}(\epsilon) \left[\mathcal{P} \frac{1}{\omega - \epsilon} - i\pi \delta(\omega - \epsilon) \right]$$

If

$$\boxed{\Gamma_{ij}^L(\omega) = \sum_{\beta} \rho_{\beta}^L(\omega) V_{i\beta}(\omega) V_{\beta j}(\omega), \quad i, j \in C} \quad (52)$$

then

$$\boxed{\Sigma_{cc}^{R(L)}(\omega) = \underbrace{\mathcal{P} \int \frac{d\epsilon}{\omega - \epsilon} \Gamma_{cc}^L(\epsilon)}_{\text{Re}} - i\pi \underbrace{\Gamma_{cc}^L(\omega)}_{\text{Im}}} \quad (53)$$

Similarly, for $\forall i, j \in C$:

$$\Sigma_{ij}^{<(L)}(\omega) = \sum_{\alpha \in L} \int d\epsilon \rho_{\alpha}^L(\epsilon) V_{i\alpha}(\epsilon) V_{\alpha j}(\epsilon) \underbrace{2\pi i f(\epsilon) \delta(\omega - \epsilon)}_{g^{<(L)}(\omega) \text{ from (12)}}$$

$$= 2\pi i f_L(\omega) \Gamma_{ij}^L(\omega), \quad \text{or} \quad \boxed{\Sigma_{cc}^{<(L)}(\omega) = 2\pi i f_L(\omega) \Gamma_{cc}^L(\omega)} \quad (54)$$

Similarly:

$$\boxed{\Sigma_{cc}^{>(L)}(\omega) = -2\pi i (1 - f_L(\omega)) \Gamma_{cc}^L(\omega)} \quad (55)$$

and

$$\boxed{\Sigma_{cc}^{A(L)}(\omega) = \mathcal{P} \int \frac{d\epsilon}{\omega - \epsilon} \Gamma_{cc}^L(\epsilon) + i\pi \Gamma_{cc}^L(\omega)} \quad (56)$$

We see that Re and Im parts of $\Sigma_{cc}^{<(L)}(\omega)$ are expressed via a single matrix function $\Gamma_{cc}^L(\omega)$ on the left lead. Its properties:

$$\Gamma_{ij}^L(\epsilon)^* = \sum_{\beta} \rho_{\beta}^L(\epsilon) V_{i\beta}^*(\epsilon) V_{\beta j}^*(\epsilon) = \sum_{\beta} \rho_{\beta}^L V_{\beta i} V_{j\beta} \equiv \Gamma_{ji}^L(\epsilon)$$

$$\boxed{\Gamma_{cc}^L(\varepsilon)^\dagger = \Gamma_{cc}^L(\varepsilon)} \leftarrow \text{Hermitian} \quad (57)$$

Therefore,

$$\left\{ \begin{array}{l} \left(\sum_{cc} <^{(L)} \right)^\dagger = - \sum_{cc} <^{(L)} \\ \left(\sum_{cc} >^{(L)} \right)^\dagger = - \sum_{cc} >^{(L)} \\ \left(\sum_{cc} R^{(L)} \right)^\dagger = \sum_{cc} A^{(L)} \\ \left(\sum_{cc} A^{(L)} \right)^\dagger = \sum_{cc} R^{(L)} \end{array} \right. \quad (58)$$

5. Current. Replace $\sum_{cc} R^{(L)}$ and $\sum_{cc} <^{(L)}$ in (48) by the above expressions (53) and (54):

$$J_L = -2e \int \frac{d\omega}{2\pi} \text{Re Tr} \left\{ \left(\int \frac{d\varepsilon}{\omega - \varepsilon} \Gamma_{cc}^L(\varepsilon) - i\pi \Gamma_{cc}^L(\omega) \right) G_{cc}^<(\omega) + 2\pi i f_L(\omega) \Gamma_{cc}^L(\omega) G_{cc}^A(\omega) \right\} \quad (59)$$

There are 3 terms:

$$\begin{aligned} \text{(i)} \quad \text{Re Tr } \Gamma_{cc}^L(\varepsilon) G_{cc}^<(\omega) &= \text{Re} \sum_{ij \in c} \Gamma_{ij}^L(\varepsilon) G_{ji}^<(\omega) = \\ &= \frac{1}{2} \sum_{ij} \left(\Gamma_{ij}^L(\varepsilon) G_{ji}^<(\omega) + \underbrace{\Gamma_{ij}^L(\varepsilon)^*}_{=\Gamma_{ji}^L(\varepsilon)} \underbrace{G_{ji}^<(\omega)^*}_{=-G_{ij}^<(\omega)} \right) = \bigcirc \\ &\quad \underbrace{- \Gamma_{ij}^L(\varepsilon) G_{ji}^<(\omega) \text{ after } i \leftrightarrow j} \end{aligned}$$

$$\text{(ii)} \quad \text{Re} \left(-i\pi \underbrace{\Gamma_{cc}^L(\omega)}_{\text{trace}} G_{cc}^<(\omega) \right) = \pi \text{Im} \left(\underbrace{\Gamma_{cc}^L(\omega) G_{cc}^<(\omega)}_{\text{trace}} \right) =$$

$$= \frac{\pi}{2i} \sum_{ij \in c} \left[\Gamma_{ij}^L(\omega) G_{ji}^<(\omega) - \Gamma_{ij}^L(\omega)^* G_{ji}^<(\omega)^* \right]$$

$$= \frac{\pi}{2i} \sum_{ij} \left(\Gamma_{ij}^L G_{ji}^< + \underbrace{\Gamma_{ji}^L G_{ij}^<}_{i \leftrightarrow j} \right) = \frac{\pi}{2} \sum_{ij} \Gamma_{ij}^L G_{ji}^< = \frac{\pi}{2} \text{Tr}(\Gamma_{cc}^L G_{cc}^<)$$

$$(iii) \text{Re} \left[2\pi i f_L(\omega) \text{Tr}(\Gamma_{cc}^L G_{cc}^A) \right] = -2\pi f_L(\omega) \text{Im} \text{Tr}(\Gamma_{cc}^L G_{cc}^A)$$

$$= -2\pi f_L \frac{1}{2i} \sum_{ij \in c} (\Gamma_{ij}^L G_{ji}^A - \Gamma_{ij}^L{}^* G_{ji}^A{}^*)$$

$$= -\frac{\pi}{i} f_L \sum_{ij} (\Gamma_{ij}^L G_{ji}^A - \underbrace{\Gamma_{ji}^L G_{ij}^A}_{i \leftrightarrow j}) = -\frac{\pi}{i} f_L(\omega) \text{Tr}[\Gamma_{cc}^L(\omega) (G_{cc}^A - G_{cc}^R)]$$

Since $\boxed{G_{ij}^R{}^* = G_{ji}^A, \quad G_{ij}^A{}^* = G_{ji}^R}$ (60)

Combine all contributions in (59):

$$\boxed{Y_L = -\frac{e}{i\hbar} \int d\omega \text{Tr} \left\{ \Gamma_{cc}^L(\omega) G_{cc}^<(\omega) + f_L(\omega) (G_{cc}^R(\omega) - G_{cc}^A(\omega)) \right\}} \quad (70)$$

Also, $\nwarrow$ (h is back)

Y_R is given by the same expression, $L \rightarrow R$, but will run in the opposite direction. We symmetrize the current:

$$\boxed{Y = \frac{1}{2} (Y_L - Y_R) = \frac{ie}{2\hbar} \int d\omega \text{Tr} \left\{ (\Gamma^L - \Gamma^R) G^< + (\Gamma_{f_L}^L - \Gamma_{f_R}^R) (G^R - G^A) \right\}} \quad (71)$$

This expression is rather general actually.

10. Use the Keldysh equation in the steady state (36):

$$G_{cc}^<(\omega) = G_{cc}^R(\omega) \underbrace{\Sigma_{cc}^<(\omega)} + G_{cc}^A(\omega) \quad (72)$$

$$\Sigma_{cc}^<(L) + \Sigma_{cc}^<(R) = 2\pi i [f_L \Gamma^L + f_R \Gamma^R] \quad (73)$$

#. With this relation $G^< \leftarrow G^A, G^R$ the whole problem becomes solved (self-contained).

11. Another expression for the current

We start from (48): ~~there are~~

$$\text{Re Tr } Y_L = -\frac{2e}{h} \int \frac{d\omega}{2\pi} \text{Re Tr} \left[\sum_{cc}^{R(L)} G_{cc}^{<} + \sum_{cc}^{<(L)} G_{cc}^A \right] \quad (48)$$

Two terms:

$$(i) \text{Re Tr} \left(\sum_{cc}^{R(L)} G_{cc}^{<} \right) = \text{Re} \sum_{ij \in c} G_{ij}^{<} \sum_{ji}^{R(L)} =$$

$$= \frac{1}{2} \sum_{ij} \left[\underbrace{G_{ij}^{<} \sum_{ji}^{R(L)} + (-G_{ji}^{<}) \left(\sum_{ji}^{A(L)} \right)}_{i \leftrightarrow j} \right] \equiv \frac{1}{2} \text{Tr} \left[G_{cc}^{<} \left(\sum_{cc}^{R(L)} - \sum_{cc}^{A(L)} \right) \right] \quad (74)$$

where

$$\sum_{cc}^{R(L)} - \sum_{cc}^{A(L)} = \frac{1}{2} \left(\int \frac{d\varepsilon}{\omega - \varepsilon} \Gamma^L(\varepsilon) - i\pi \Gamma^L(\omega) \right) -$$

$$- \frac{1}{2} \left(\int \frac{d\varepsilon}{\omega - \varepsilon} \Gamma^L(\varepsilon) + i\pi \Gamma^L(\omega) \right) = -\frac{1}{2} 2\pi i \Gamma^L(\omega) \quad (75)$$

$$\equiv \sum_{cc}^{>} - \sum_{cc}^{<} \quad (76)$$

(see (53)-(56)).

$$(ii) \text{Re Tr} \left(G_{cc}^A \sum_{cc}^{<(L)} \right) = \frac{1}{2} \sum_{ij \in c} \left[G_{ij}^A \sum_{ji}^{<(L)} + \underbrace{G_{ji}^A \left(-\sum_{ji}^{<(L)} \right)}_{i \leftrightarrow j} \right]$$

$$= \frac{1}{2} \sum_{ij} (G_{ij}^A - G_{ij}^R) \sum_{ji}^{<(L)} \equiv \frac{1}{2} \text{Tr} \left[(G^{<} - G^{>}) \Sigma^{<} \right] \quad (77)$$

Collecting both results (75)-(77) in (48), gives:

$$Y_L = -\frac{2e}{h} \int \frac{d\omega}{2\pi} \underbrace{\left[G^{<} (\Sigma^{>} - \Sigma^{<}) + (G^{<} - G^{>}) \Sigma^{<} \right]}_{G^{<} \Sigma^{>} - \cancel{G^{<} \Sigma^{<}} + \cancel{G^{>} \Sigma^{<}} - G^{>} \Sigma^{<}}$$

$$\hookrightarrow \boxed{J_L = + \frac{e}{\hbar} \int \frac{d\omega}{2\pi} \text{Tr} [G_{cc}^>(\omega) \Sigma_{cc}^{< (4)}(\omega) - G_{cc}^<(\omega) \Sigma_{cc}^{> (4)}(\omega)]} \quad (78)$$

which is also rather exact.

12. Final formula for the TB Hamiltonian.

From the Keldysh equations: (72):

$$G_{cc}^* = 2\pi i G_{cc}^R (f_L \Gamma_{cc}^L + f_R \Gamma_{cc}^R) G_{cc}^A$$

$$G_{cc}^* = -2\pi i G_{cc}^R [(1-f_L) \Gamma_{cc}^L + (1-f_R) \Gamma_{cc}^R] G_{cc}^A$$

$$\Sigma_{cc}^{< (4)}(\omega) = 2\pi i f_L \Gamma_{cc}^L, \quad \Sigma_{cc}^{> (4)}(\omega) = -2\pi i (1-f_L) \Gamma_{cc}^L(\omega)$$

Combining all these in (78), we get:

$$\text{Tr}[\dots] = (2\pi i)^2 \text{Tr} \{ G^R [-(1-f_L) \Gamma^L - (1-f_R) \Gamma^R] G^A f_L \Gamma^L - \\ - G^R (f_L \Gamma^L + f_R \Gamma^R) G^A \cdot (-(1-f_L)) \Gamma^L \}$$

$$= -(2\pi)^2 \text{Tr} \{ \cancel{G^R \Gamma^L G^A \Gamma^L} (1-f_L) f_L - G^R \Gamma^R G^A \Gamma^L (1-f_R) f_L \\ + f_L (1-f_L) \cancel{G^R \Gamma^L G^A \Gamma^L} + (1-f_L) f_R G^R \Gamma^R G^A \Gamma^L \}$$

$$= -(2\pi)^2 \cancel{\mathbb{I}} [-f_L(1-f_R) + f_R(1-f_L)] \text{Tr}(G^R \Gamma^R G^A \Gamma^L)$$

$$\underbrace{-f_L + f_L f_R + f_R - f_R f_L}_{-f_L + f_L f_R + f_R - f_R f_L = f_R - f_L}$$

$$= (2\pi)^2 (f_L - f_R) \text{Tr}(G^R \Gamma^R G^A \Gamma^L)$$

$$\hookrightarrow \boxed{J_L = \frac{2\pi e}{\hbar} \int d\omega (f_L(\omega) - f_R(\omega)) \text{Tr}(G^R \Gamma^R G^A \Gamma^L)_c} \quad (79)$$

Define the transmission coefficient (matrix):

$$T_{cc}(\omega) = (\Gamma_L^c)^{1/2} G_{cc}^{LR} (\Gamma_R^c)^{1/2} \quad (80)$$

then

$$\begin{aligned} \text{Tr}(G^R \Gamma_R G^A \Gamma_L) &= \text{Tr}(G^R (\Gamma_R)^{1/2} (\Gamma_R)^{1/2} G^A (\Gamma_L)^{1/2} (\Gamma_L)^{1/2}) \\ &= \text{Tr} \left[\underbrace{(\Gamma_L)^{1/2} G^R (\Gamma_R)^{1/2}}_{T(\omega)} \cdot \underbrace{(\Gamma_R)^{1/2} G^A (\Gamma_L)^{1/2}}_{T^\dagger(\omega) \text{ as } (G^A)^\dagger = G^R} \right] \end{aligned}$$

$$\hookrightarrow \boxed{J_L = \frac{2\pi e}{h} \int d\omega [f_L - f_R] \text{Tr}[T(\omega) T^\dagger(\omega)]} \quad (81)$$

(Landauer form).

13. Self-consistency

The electrostatic potential inside the junction $\phi(x)$ enters $\mu(x)$ in all parts of the Hamiltonian. It depends on the density of electrons and the boundary conditions.

(i) Density of electrons:

$$\rho(r) = \langle \psi_{\mathbf{r}}^\dagger(x,t) \psi_{\mathbf{r}}(x,t) \rangle = -i \underbrace{G^<(x,t, x,t)}_{\sum_{ij} G_{ij}^<(t,t) N_i(x) N_j^*(x)}$$

$$\hookrightarrow \rho_{cc} = -i G_{cc}^<(t,t) = -i \int \frac{d\omega}{2\pi} G_{cc}^<(\omega) \quad (82)$$

$$G_{cc}^<(\omega) = G_{cc}^R(\omega) \left(\underbrace{\sum_{cc}^<(A)}_{2\pi i f_L \Gamma_{cc}^L(\omega)} + \underbrace{\sum_{cc}^<(L)}_{2\pi i f_R \Gamma_{cc}^R(\omega)} \right) G_{cc}^A(\omega) \quad (83)$$

and hence can be calculated (as G^R and G^A are self-contained).

(ii) Boundary conditions

ρ_{cc} is only given within C region. We have to solve the Poisson eq. for the Hartree potential

$$\Delta V_H(\vec{r}) = -4\pi \rho(\vec{r}) \quad (84)$$

to be used in the SCF equations. If $\phi(\vec{r})$ is the potential due to $\rho(\vec{r})$, i.e. $\Delta\phi(\vec{r}) = -4\pi \rho(\vec{r})$, then generally

$$V_H(\vec{r}) = \phi(\vec{r}) + \vec{\alpha} \cdot \vec{r} + \beta, \quad \vec{r} \in C \quad (85)$$

and $\vec{\alpha}$ and β are to be determined from the boundary conditions. Along $\parallel$ directions ($\perp$ to the transport direction)

$\vec{\alpha}_{\parallel} = 0$, so we have two constants $\alpha_{\perp}$ and β . These are obtained by calculating the electrostatic $V_H(\vec{r})$ out the C-R and L-C boundaries, V_H^{CR} and V_H^{LC} , from an independent calculation (eg. bulk L and bulk R), and then shifting one of them by V (the bias):

$$V_H(\vec{r}) = \phi(\vec{r}) + \frac{V + (V_L - V_R)}{z_R - z_L} z + \frac{V_R z_L - (V + V_L) z_R}{z_R - z_L} \quad (86)$$

where: $V_L \equiv -V_H^{LC} + \phi_L$, $V_R \equiv -V_H^{CR} + \phi_R$ (and are both ~~different~~ still functions of $\vec{r}_{\parallel}$), z_L and z_R are the z -boundaries of the C

Region:

